الجمهورية اللبنانية
وزارة التربية والتعليم العالي
المديرية العامة للتعليم المهني والتقني
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الامتحانات الرسمية لشهادة الامتياز الفني
لدورة عام 2021 الاولى

الاسم والشهرة: ----------
رقم المرشح: ----------

الاختصاص والرمز : الالكترونيك (20203)
المادة : الالكترونيك الرقمي
المستندات المسموح بها : آلة حاسبة غير مبرمجة

المعدل : (10)
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الإجابات على "كراس الإجابة"

A- Choose the correct answer : (1pt/correct answer)

1- If the register TRISB = $(0000\ 0000)_B$, then PORTB will be totally initiated as :

a- OUTPUT
b- HALF INPUT – HALF OUTPUT
c- INPUT
d- NOTHING

2- If $(0x0D) = (AA)_H$ and the instruction [COMF 0x0D, F] is executed, the result will be:

a- $W = (55)_H$
b- $W = (54)_H$
c- $(0x0D) = (55)_H$
d- $(0x0D) = (54)_H$

3- The result if the instruction [ANDLW 0xAA] with $W = (FF)_H$ will be:

a- $W = (FF)_H$
b- $W = (FA)_H$
c- $W = (AA)_H$
d- $W = (00)_H$

4- To change to BANK1, we write:

a- RP1 = 0 and RP0 = 1
b- RP1 = 1 and RP0 = 0
c- RP1 = 1 and RP0 = 1
d- RP1 = 0 and RP0 = 0

5- To turn ON a LED connected to PIN RB3 of PORTB, we write:

a- BCF PORTB, 3
b- BSF PORTB, 3
c- BSF TRISB, 3
d- BCF TRISB, 3

6- If the register DDRA = $(0000\ 0000)_B$, then PORTA will be totally initiated as :

a- OUTPUT
b- HALF INPUT – HALF OUTPUT
c- INPUT
d- NOTHING

7- If $DDRB = (0F)_H$, then port B will be initiated:

a- All as Output
b- Half Input – Half Output
c) All as Input
d- Nothing

8- The result if the instruction [ANDB #$AA] with $B = (FF)_H$ will be:

a- $B = (FF)_H$
b- $B = (FA)_H$
c- $B = (AA)_H$
d- $B = (00)_H$

9- To achieve the difference $(143 - 35)_D$ with $B = (143)_D$, we write the following instruction:

a- SUBB #$23
b- SUBB #$35
c- SUBB #$22
d- SUBB $0035

10- To turn ON a LED connected to PIN PB3 of PORTB, we write LDB #$08 and:

a- STA PORTB
b- STA ORB
c- STA ORA
d- STB ORB

Micro 2021 - 1st

Part A. Multiple choise

Answer.	Discussion
1. a.	TRISB = 0x00 → Output = 0xFF → Input
2. C AA = 1010 1010 COMAA = 01010101 = 55	COMF 0x0D, F, will complement the contents of file (0x0D) & "F" will put the value in 0x0D not W. COM AA → 55
3. C	AA = 1010 1010 FF = 1111 1111 2 AND 1010 1010 = AA
4. a.	RP1 is not emplemented & should remain Zero. RP0 = {0 = Bank0 1 = Bank1.
5. b	BSF PortB,3 "Bit set file will make bit3 = "1"
6. C	in 6809, DDRB = 0's ⇒ input = 1's ⇒ output.
7. b.	
8. C.	AA 10101010 FF 11111111 2 AND 10101010
9. a.	SUBB #$23 with subtract (35)d from B. $(23)_{16} = (2\times16^1)+(3\times16^0) = 32+3 = 35$.
10 d. STB ORB	because 08 is stored in B not A.

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B - Treat 2 questions only : (5 pts / question)

I- Write a program that sequentially outputs at ***port B*** of a PIC16F84A, the bytes of the below table with a certain delay of your choice between two consecutive outputs:

RB7	RB6	RB5	RB4	RB3	RB2	RB1	RB0
1	0	0	0	0	0	0	1
0	1	0	0	0	0	1	0
0	0	1	0	0	1	0	0
0	0	0	1	1	0	0	0
0	0	0	1	1	0	0	0
0	0	1	0	0	1	0	0
0	1	0	0	0	0	1	0
1	0	0	0	0	0	0	1

PART B.

1)

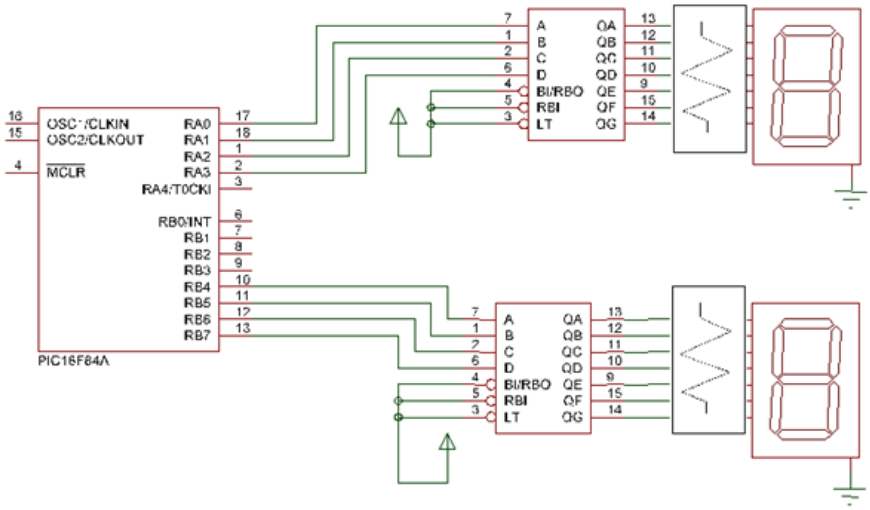
```
#include (pic16F84A.inc)
num1  equ 0x20
num2  equ 0x21
org 0x00
bsf STATUS, RP0
clrf TRISB
bcf STATUS, RP0

START
   MOVLW 0x80
   MOVWF num1
   MOVLW 0x01
   MOVWF num2.
   CLRW
```

```
main
IORWF num1
IORWF num2
MOVWF PORTB
RRF num1
RLF num2
NOP
NOP
btfss STATUS,C
goto main
goto START
END
```

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II- Consider the following circuit built around a PIC16F84A :



Write the program to build a decimal ODD counter from 01 to 99. PORTA is for tens and PORTB is for units. Make a delay of your choice between two consecutive displays.

2)

```
#include (PIC16F84A.INC)
Count1   equ 0x20
Count2   equ 0x21

org 0x00
bsf STATUS, RP0
clrf TRISA
clrf TRISB
bcf STATUS, RP0
START
Clrf Count1
Incf Count1

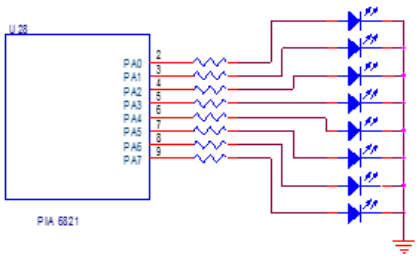
Clrf Count2
Clr PORTA
Clr PORTB

Main
Nop
NOP
 swapf Count1,W
 MOVWF PORTB
 INCf Count1
 Incf Count1

 BTFSC Count1,3
 BTFSS Count1,1
 goto main
 Clrf Count1
 Incf Count1
 Incf Count2
 MOVF Count2,0
 MOVWF PORTA
 BTFSC Count2,3
 BTFSS Count2,1
  goto main
  goto START
   END
```

III- Write the program that displays sequentially at port A of a PIA6821 connected to MP6809, the bytes of the following table, with a certain delay of your choice between two consecutive outputs.

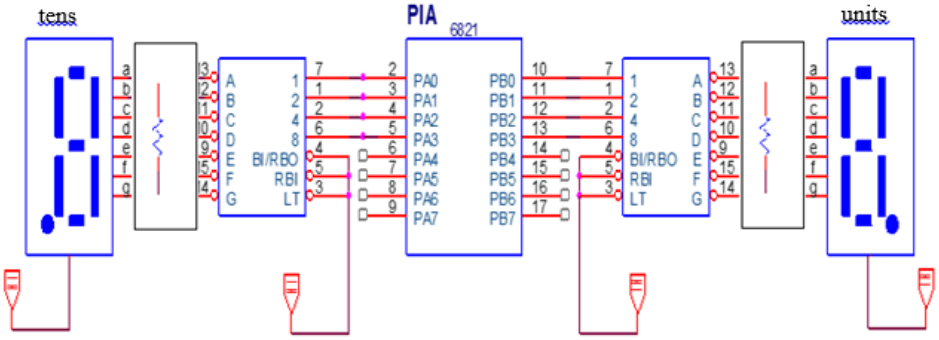
PA7	PA6	PA5	PA4	PA3	PA2	PA1	PA0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	0
0	1	0	0	0	1	0	0
1	0	0	0	1	0	0	0
1	0	0	0	1	0	0	0
0	1	0	0	0	1	0	0
0	0	1	0	0	0	1	0
0	0	0	1	0	0	0	1



3)

```
DDRA  EQU $8000
ORA   EQU $8000
CRA   EQU $8001
      ORG $FC00
      CLR CRA
      CLR DDRA
      COM DDRA
      LDA #$04
      STA CRA.
START:
      LDA #$11
LOOP1:
      STA PORTA
      JSR Delay.
      ROLA
      CMPA #$10
      BNE LOOP1
      RORA
LOOP2:
      STA PORTA
      JSR Delay.
      RORA
      BCC LOOP2
      BRA START
      END
Delay:
      CLRB
LOOP3:
      INCB
      BCC LOOP3
      RTS
```

IV- By referring to the below circuit, write the program to build a counter from ***00*** to ***99***.




```
DDRA EQU $8000
ORA EQU $8000
CRA EQU $8001
DDRB EQU $8002
ORB EQU $8002
CRB EQU$8003
ORG $FC00
CLR CRA
CLR CRB
CLR DDRA
COM DDRA
CLR DDRB
COM DDRB
LDA #$04
STA CRA
STA CRB
START:
        CLR ORA
        CLR ORB
LOOP:
        INC ORB
        LDA ORB
        CMPA #$09
        BNE LOOP
        CLR ORB
        INC ORA
        LDA ORA
        CMPA #$09
        BNE LOOP
        BRA START
        END
```

ANNEXE / APPENDIX

Instruction set of 6809

ADCA, ADCB, ADDA, ADDB, ADDD, ANDA, ANDB, ANDCC, ASLA, ASLB, ASL, ASRA, ASRB, ASR, BITA, BITB, CLRA, CLRB, CLR, CMPA, CMPB, CMPD, CMPU, CMPS, CMPX, CMPY, COMA, COMB, COM, CWAI, DAA, DECA, DECB, DEC, EORA, EORB, EXG, INCA, INCB, INC, JMP, JSR, LDA, LDB, LDD, LDS, LDU, LDX, LDY, LEAS, LEAU, LEAX, LEAY, LSLA, LSLB, LSL, LSRA, LSRB, LSR, MUL, NEGA, NEGB, NEG, NOP, ORA, ORB, ORCC, PSHS, PSHU, PULS, PULU, ROLA, ROLB, ROL, RORA, RORB, ROR, RTI, RTS, SBCA, SBCB, SEX, STA, STB, STD, STS, STU, STX, STY, SUBA, SUBB, SUBD, SWI, SWI2, SWI3, SYNC, TFR, TSTA, TSTB, TST.

PIA:

ORA et DDRA = $8000	**CRA = $8001**
ORB et DDRB = $8002	**CRB = $8003**

Instruction set of PIC 16F84A

ADDWF, ANDWF, CLRF, CLRW, COMF, DECF, DECFSZ, INCF, INCFSZ, IORWF, MOVF, MOVWF, NOP, RLF, RRF, SUDWF, SWAPF, XORWF, BCF, BSF, BTFSC, BTFSS, ADDLW, ANDLW, CALL, CLRWDT, GOTO, IORLW, MOVLW, RETFIE, RETLW, RETURN, SLEEP, SUBLW, XORLW.

Internal architecture of PIC16F84A

File Address			File Address
00h	Indirect addr.(1)	Indirect addr.(1)	80h
01h	TMR0	OPTION_REG	81h
02h	PCL	PCL	82h
03h	STATUS	STATUS	83h
04h	FSR	FSR	84h
05h	PORTA	TRISA	85h
06h	PORTB	TRISB	86h
07h	—	—	87h
08h	EEDATA	EECON1	88h
09h	EEADR	EECON2(1)	89h
0Ah	PCLATH	PCLATH	8Ah
0Bh	INTCON	INTCON	8Bh
0Ch	General Purpose Registers (SRAM)	Mapped (accesses) in Bank 0	8Ch
			FFh
	Bank 0	Bank 1	

PS2:PS0: Prescaler Rate Select bits

Bit Value	TMR0 Rate	WDT Rate
000	1 : 2	1 : 1
001	1 : 4	1 : 2
010	1 : 8	1 : 4
011	1 : 16	1 : 8
100	1 : 32	1 : 16
101	1 : 64	1 : 32
110	1 : 128	1 : 64
111	1 : 256	1 : 128